M3503-14 WEEK 3 2020/21

HIGHER ORDER LINEAR DIFFERENTIAL EQUATIONS

HOMOGENEOUS LINEAR EQUATION WITH CONSTANT COEFFICIENTS

An nth order differential equation of the form

$$a_n\frac{d^n y}{dx^n} + a_{n-1}\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_2\frac{d^2y}{dx^2} + a_1\frac{dy}{dx} + a_0 y = 0 \quad \text{—①}$$

is said to be homogeneous while

$$a_n\frac{d^n y}{dx^n} + a_{n-1}\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_1\frac{dy}{dx} + a_0 y = f(x) \text{ with } f(x)\neq 0$$

is nonhomogeneous.

Consider a second order homogeneous equation

$$a_2 y'' + a_1 y' + a_0 y = 0 \quad \text{—②}.$$

Looking at eq^n ②, we notice that its solution has the property that its 2nd derivative can be expressed as a linear combination of its first and zeroth derivative. One such function is $y = e^{rx}$, substituting into eq^n ② we get

$$a_2 r^2 e^{rx} + a_1 r e^{rx} + a_0 e^{rx} = 0$$

$$\Rightarrow e^{rx}(a_2 r^2 + a_1 r + a_0) = 0$$

$$\Rightarrow a_2 r^2 + a_1 r^2 + a_0 = 0 \quad \text{—③}$$

Thus $y = e^{rx}$ is a solution to eq^n ② if r satisfies eq^n ③, which is known as the auxiliary equation or the characteristic equation associated with eq^n ②.

From eqn ③

$$r_1 = \frac{-a_1 + \sqrt{a_1^2 - 4a_2a_0}}{2a_2} \quad \text{and} \quad r_2 = \frac{-a_1 - \sqrt{a_1^2 - 4a_1a_0}}{2a_2}$$

There are 3 possibilities

① We have distinct real roots
② we have repeated roots
③ we have complex root.

We will demonstrate later with examples, how a solution is presented in each case.

Superposition Principle For Homogeneous Linear Equations

Let $y_1, y_2, \ldots, y_k$ be solutions to eqn ①, then
$y = c_1y_1 + c_2y_2 + \cdots + c_ky_k$, where c_i's are constants, is also a solution to eqn ①.

Linear Dependence / Independence

A set of functions $y_1, y_2, \ldots, y_n$ is linearly dependent on an interval I if there exist constants $c_1, c_2, \ldots, c_n$, not all zeros, such that

$$c_1y_1 + c_2y_2 + c_3y_3 + \cdots + c_ny_n = 0$$

on the interval I, otherwise, it is independent.

For the solution of differential equations, we are mostly interested in linearly independent solutions, also known as a fundamental set of solutions. To determine whether a set of solutions is linearly independent, we use the following determinant

Suppose each of the functions $y_1, y_2, \ldots, y_n$ has at least $n-1$ derivatives, the determinant

$$W(y_1, y_2, \ldots, y_n) = \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & & y_n' \\ \vdots & \vdots & & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{vmatrix}$$

is called the Wronskian of the functions

Criterion For Linearly Independent Solutions

Let $y_1, y_2, \ldots y_n$ be solutions to an nth order homogeneous linear equation on an interval I. The set of solutions is linearly independent on I if $W(y_1, y_2, \ldots, y_n) \neq 0 \ \forall x$ in I

N.B $W(y_1, y_2, \ldots, y_n)$ is either zero or non-zero on I.
Thus if it can be shown that for some $x_0 \in I$ $W(y_1, y_2, \ldots y_n) \neq 0$, then the set of functions $y_1, y_2, \ldots y_n$ is linearly independent on I.

General Solution Of Homogeneous Linear Equation

Let $y_1, y_2, \ldots, y_n$ be a fundamental set of solutions of an nth order homogeneous linear equation on interval I. The general solution of the equation on the interval is

$$y = C_1 y_1 + C_2 y_2 + \cdots + C_n y_n$$

Where C_i's are arbitrary constants.

As mentioned earlier, we may have distinct real roots, repeated roots or complex roots.
Here is how the general solution is presented in each case.

Distinct Real Roots

If the auxiliary equation has distinct real roots r_1 and r_2, then $y_1 = e^{r_1 x}$ and $y_2 = e^{r_2 x}$ are both solution to the homogeneous equation and $y = C_1 e^{r_1 x} + C_2 e^{r_2 x}$ is the general solution

Examples
Solve ① $y'' - 9y = 0$
② $y''' + y'' - 6y' + 4y = 0$

Soln ①. The auxiliary equation is

$$r^2 - 9 = 0$$
$$\Rightarrow r = \pm 3$$

∴ The general solution is

$$y = C_1 e^{3x} + C_2 e^{-3x}$$

②. The auxiliary eqn is

$$r^3 + r^2 - 6r + 4 = 0$$
$$\Rightarrow (r-1)(r^2 + 2r - 4) = 0$$
$$\Rightarrow r = 1, \text{ or } r = -1 + \sqrt{5} \text{ or } r = -1 - \sqrt{5}$$

∴ The general equation is

$$y = C_1 e^{x} + C_2 e^{(-1+\sqrt{5})x} + C_3 e^{(-1-\sqrt{5})x}$$

Repeated Roots

If the auxiliary equation has repeated roots, both $y_1 = e^{rx}$ and $y_2 = xe^{rx}$ are solutions to the associated homogeneous equation and $y = C_1e^{rx} + C_2xe^{rx}$ is the general solution.

Examples

Solve ①. $y'' + 6y' + 9y = 0$

② . $y''' + 3y'' - 4y = 0$

Soln

①. The auxiliary equation is

$$r^2 + 6r + 9 = 0$$

$$\Rightarrow (r+3)^2 = 0$$

$$\Rightarrow r = -3$$

∴ The general solution is

$$y = C_1e^{-3x} + C_2xe^{-3x}$$

②. The auxiliary equation is

$$r^3 + 3r^2 - 4 = 0$$

$$\Rightarrow (r-1)(r^2 + 4r + 4) = 0$$

$$\Rightarrow (r-1)(r+2)^2 = 0$$

$$\Rightarrow r = 1 \text{ or } r = -2$$

∴ The general solution is

$$y = C_1e^{x} + C_2e^{-2x} + C_3xe^{-2x}$$

Complex Roots

If the auxiliary equation has complex roots $r_1 = \alpha + i\beta$ and $r_2 = \alpha - i\beta$, then both $y_1 = e^{\alpha x}\cos\beta x$ and $y_2 = e^{\alpha x}\sin\beta x$ are solutions to the homogeneous equation and the general solution is

$$y = C_1 e^{\alpha x}\cos\beta x + C_2 e^{\alpha x}\sin\beta x$$

where C_1 and C_2 are arbitrary constant.

Examples

Solve ① $y'' - 4y' + 7y = 0$

② $y''' + y'' + 3y' - 5y = 0$

Soln ① The auxiliary equation is

$$r^2 - 4r + 7 = 0$$

$$r = \frac{4 \pm \sqrt{16 - 4\times 7}}{2}$$

$$= 2 \pm \sqrt{-3} \quad = 2 \pm i\sqrt{3}$$

$\therefore y = C_1 e^{2x}\cos\sqrt{3}x + C_2 e^{2x}\sin(\sqrt{3}x)$

② The auxiliary equation is

$$r^3 + r^2 + 3r - 5 = 0$$

$$(r-1)(r^2 + 2r + 5) = 0$$

$r = 1$ or $r = \frac{-2 \pm \sqrt{4 - 20}}{2}$

$$= -1 \pm i2$$

$\therefore y = C_1 e^{x} + C_2 e^{-x}\cos 2x + C_3 e^{-x}\sin 2x$

Nonhomogeneous Linear Equations With Constant Coefficients

These are equations of the form

$$a_n\frac{d^n y}{dx^n} + a_{n-1}\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_1 y' + a_0 y = f(x) \quad \text{with} \quad f(x) \neq 0$$

There are different methods that can be used to solve nonhomogeneous equation. One such method is the

1. Undetermined Coefficient Method

To find the general solution using this method, we find the complementary solution y_c. y_c is the solution of the associated homogeneous equation

$$a_n\frac{d^n y}{dx^n} + a_{n-1}\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_1 y' + a_0 y = 0$$

We also guess the form of the particular solution y_p of the equation by looking at the right hand side of the equation $f(x)$ and noting the roots of the associated auxiliary equation. After identifying the form of y_p, we solve for undetermined coefficients. The general solution is given by

$$y = y_c + y_p.$$

The method of undetermined coefficients can only be used if the right hand side of the equation $f(x)$ has the property that all its derivatives can be written in terms of a finite number of functions. Functions with this property are polynomials, exponentials, Sines Cosines, and the products of these functions. Also it only works for linear equations with constant coefficients.

Here is a summarized guide on how to choose the form of the particular solution y_p

- If $f(x) = Cx^m e^{rx}$, where m is a nonnegative integer, we choose

$$y_p = x^s(A_m x^m + \dots + A_1 x + A_0)e^{rx}$$

with * $s=0$ if r is not the root of the associated auxiliary equation.

* $s=1$ if r is a simple real root of the associated auxiliary equation.

* $s=2$ if r is the repeated real root of the associated auxiliary equation

- If $f(x) = \begin{cases} Cx^m e^{\alpha x}\cos\beta x \\ Cx^m e^{\alpha x}\sin\beta x \end{cases}$ with $\beta \neq 0$, then

$$y_p = x^s(A_m x^m + \dots + A_1 x + A_0)e^{\alpha x}\cos\beta x + x^s(B_m x^m + \dots + B_1 x + B_0)e^{\alpha x}\sin\beta x$$

with * $s=0$ if $\alpha + i\beta$ is not a root to the associated auxiliary equation

* $s=1$ if $\alpha + i\beta$ is a root (complex root) to the associated auxiliary equation

Example 1

Find the particular solution to

(a) $y'' + 2y' - y = 10$

(b) $y'' + y' = 2$

Solⁿ

(a) Let $y_p = A$

$\Rightarrow y_p' = 0$

$\Rightarrow y_p'' = 0$

Substituting into the equation we get

$-A = 10 \Rightarrow A = -10$

$\therefore y_p = -10$

(b) Here notice that if we choose $y_p = A$, both first and second derivatives are zeros, so their sum is not 2 but zero. Thus lets try

$y_p = Ax$

$\Rightarrow y_p' = A$

$\Rightarrow y_p'' = 0$

Substituting we get

$A = 2$

$\therefore y_p = 2x$

N.B we multiplies by x because $z = 2e^{0 \cdot x}$ and 0 is the root of the associated auxiliary equation

Example 2.

Find the particular solution to

(a) $2y' + y = 3x^2$

(b) $y'' + 4y' - 2y = 2x^3 - 3x + 6$

Soln

(a) Let $y_p = Ax^2 + Bx + C$

$\Rightarrow y_p' = 2Ax + B$

Substituting we get

$4Ax + 2B + Ax^2 + Bx + C = 3x^2$

$\Rightarrow A = 3$, $4A + B = 0$ and $2B + C = 0$

$\Rightarrow B = -12$ and $C = 24$

$\therefore y_p = 3x^2 - 12x + 24$

(b) Let $y_p = Ax^3 + Bx^2 + Cx + D$

$\Rightarrow y_p' = 3Ax^2 + 2Bx + C$

$\Rightarrow y_p'' = 6Ax + 2B$

Substituting we get

$6Ax + 2B + 12Ax^2 + 8Bx + 4C - 2Ax^3 - 2Bx^2 - 2Cx - 2D = 2x^3 - 3x + 6$

$-2A = 2 \Rightarrow A = -1$

$12A - 2B = 0 \Rightarrow B = -6$

$6A + 8B - 2C = -3 \Rightarrow C = -\frac{51}{2}$

$2B + 4C - 2D = 6 \Rightarrow D = -60$

$y_p = -x^3 - 6x^2 - \frac{51}{2}x - 60$

Example 3

Find the particular solution to

(a) $y'' + y = 2^x$

(b) $y'' - 5y' + 4y = 8e^x$

sol^n (a) Let $y_p = A2^x$

$\Rightarrow y_p' = A(\ln 2)\, 2^x$

$\Rightarrow y_p'' = A(\ln 2)^2\, 2^x$

substituting

$$A(\ln 2)^2\, 2^x + A2^x = 2^x$$

$$\Rightarrow A(\ln 2)^2 + A = 1$$

$$\Rightarrow A = \frac{1}{(\ln 2)^2 + 1}$$

$$\therefore y_p = \frac{2^x}{(\ln 2)^2 + 1}$$

(b) Notice that 1 is the root of the auxiliary equation, thus we choose

$y_p = Axe^x$ instead of Ae^x

$\Rightarrow y_p' = Axe^x + Ae^x$

$\Rightarrow y_p'' = Axe^x + 2Ae^x$

Substituting, we get

$$Axe^x + 2Ae^x - 5Axe^x - 5Ae^x + 4Axe^x = 8e^x$$

$$\Rightarrow -3Ae^x = 8e^x$$

$$\Rightarrow A = -\frac{8}{3}$$

$$\therefore y_p = -\frac{8}{3}xe^x$$

Example 4

Find the particular solution to

(a) $y'' - y' + y = 2\sin 3x$

(b) $y''' - 2y'' + y = \cos x$

(c) $y'' + y = \cos x$

Soln

(a) Let $y_p = A\sin 3x + B\cos 3x$

$\Rightarrow y_p' = A\cos 3x - 3B\sin 3x$

$\Rightarrow y_p'' = -9A\sin 3x - 9B\cos 3x$

Substituting we get

$-9A\sin 3x - 9B\cos 3x - 3A\cos 3x + 3B\sin 3x + A\sin 3x + 3\cos 3x = 2\sin 3x$

$\Rightarrow 8A\sin 3x + 3B\sin 3x - 8B\cos 3x - 3A\cos 3x = 2\sin 3x$

$\Rightarrow 8A + 3B = 2$ and $-8B - 3A = 0$

$\Rightarrow A = -\frac{8}{3}B$

Substituting for A we get

$\frac{64}{3}B + 3B = 2$

$\Rightarrow B = \frac{6}{73}$, so $A = -\frac{16}{73}$

$\therefore y_p = -\frac{16}{17}\sin 3x + \frac{6}{73}\cos 3x$

(b) Let $y_p = A\cos x + B\sin x$

$\Rightarrow y_p' = -A\sin x + B\cos x$

$\Rightarrow y_p'' = -A\cos x - B\sin x$

$\Rightarrow y_p''' = A\sin x - B\cos x$

Substituting

$A\sin x - B\cos x + 2A\cos x + 2B\sin x + A\cos x + B\sin x = \cos x$

$\Rightarrow A + 2B + B = 0$ and $-B + 2A + A = 1$

$\Rightarrow A = -3B$

Substituting for A we get

$-B - 9B = 1$

$\Rightarrow B = -\frac{1}{10}$ and $A = \frac{3}{10}$

$\therefore y_p = \frac{3}{10}\cos x - \frac{1}{10}\sin x$

c) Let $y_p = Ax\cos x + Bx\sin x$ N.B: i is the root to the associated auxiliary equation

$y_p' = -Ax\sin x + A\cos x + Bx\cos x + B\sin x$

$y_p'' = -Ax\cos x - 2A\sin x - Bx\sin x + 2B\cos x$

Substituting

$-Ax\cos x - 2A\sin x - Bx\sin x + 2B\cos x + Ax\cos x + Bx\sin x = \cos x$

$\Rightarrow A = 0$ and $B = \frac{1}{2}$

$\therefore y_p = \frac{1}{2}x\sin x$

Example 5

Find the particular solution to

(a) $y''' + y'' - 2y = xe^x$

(b) $y'' + y = x\cos x$

c) $y'' - 2y' + 5y = e^x\cos 2x$

d) $y'' - 2y = x\sin(x)e^x$

Soln (a) Let $y_p = (Ax^2+Bx)e^x$

$\Rightarrow y_p' = (Ax^2+Bx)e^x + (2Ax+B)e^x$

$\Rightarrow y_p'' = (Ax^2+Bx)e^x + 2(2Ax+B)e^x + 2Ae^x$

$\Rightarrow y_p''' = (Ax^2+Bx)e^x + 3(2Ax+B)e^x + 6Ae^x$

Substituting we get

$(Ax^2+Bx)e^x + 3(2Ax+B)e^x + 6Ae^x + (Ax^2+Bx)e^x + 2(2Ax+B)e^x + 2Ae^x$
$- 2(Ax^2+Bx)e^x = xe^x$

$\Rightarrow 5(2Ax+B)e^x + 8Ae^x = xe^x$

$\Rightarrow 10A = 1$ and $5B+8A = 0$

$\Rightarrow A = \frac{1}{10}$ and $B = -\frac{4}{25}$

$\therefore y_p = \left(\frac{x^2}{10} - \frac{4}{25}x\right)e^x$

(b) Let $y_p = (Ax^2+Bx)\cos x + (Dx^2+Cx)\sin x$

$\Rightarrow y_p' = -(Ax^2+Bx)\sin x + (2Ax+B)\cos x + (Dx^2+Cx)\cos x +$
$(2Dx+C)\sin x$

$\Rightarrow y_p'' = -(Ax^2+Bx)\cos x - 2(Ax+B)\sin x + 2A\cos x$
$- (Dx^2+Cx)\sin x + 2(2Dx+C)\cos x + 2D\sin x$

Substituting we get

$-2(Ax+B)\sin x + 2A\cos x + 2(2Dx+C)\cos x + 2D\sin x = x\cos x$

So we have

$4D = 1 \Rightarrow D = \frac{1}{4}$

$-4A = 0 \Rightarrow A = 0$

$2A+2C = 0 \Rightarrow C = 0$

$-2B+2D = 0 \Rightarrow B = D = \frac{1}{4}$

$\therefore y_p = \frac{1}{4}x\cos x + \frac{1}{4}x^2\sin x$

c) Let $y_p = Axe^x \cos 2x + Bxe^x \sin 2x$

$\Rightarrow y_p' = Axe^x \cos 2x + Ae^x \cos 2x - 2Axe^x \sin 2x + Bxe^x \sin 2x + Be^x \sin 2x + 2Bxe^x \cos 2x$

$\Rightarrow y_p'' = Axe^x \cos 2x + 2Ae^x \cos 2x - 4Axe^x \sin 2x - 4Ae^x \sin 2x - 4Axe^x \cos 2x + Bxe^x \sin 2x + 2Be^x \sin 2x + 4Bxe^x \cos 2x + 4Be^x \cos 2x - 4Bxe^x \sin 2x$

$= -3Axe^x \cos 2x + 2Ae^x \cos 2x - 4Axe^x \sin 2x - 4Ae^x \sin 2x - 3Bxe^x \sin 2x + 2Be^x \sin 2x + 4Bxe^x \cos 2x + 4Be^x \cos 2x$

Substituting we get

$-3Axe^x \cos 2x + 2Ae^x \cos 2x - 4Axe^x \sin 2x - 4Ae^x \sin 2x - 3Bxe^x \sin 2x + 2Be^x \sin 2x + 4Bxe^x \cos 2x + 4Be^x \cos 2x - 2Axe^x \cos 2x - 2Ae^x \cos 2x + 4Axe^x \sin 2x - 2Bxe^x \sin 2x - 2Be^x \sin 2x - 4Bxe^x \cos 2x + 5Axe^x \cos 2x + 5Bxe^x \sin 2x = e^x \cos 2x$

$\Rightarrow -4Ae^x \sin 2x + 4Be^x \cos 2x = e^x \cos 2x$

$\Rightarrow -4A = 0 \Rightarrow A = 0$

and $4B = 1 \Rightarrow B = \frac{1}{4}$

$\therefore y_p = \frac{1}{4} xe^x \sin 2x$

d) let $y_p = (Ax+B)\sin(x)e^x + (Dx+C)\cos(x)e^x$

$\Rightarrow y_p' = (Ax+B)\sin(x)e^x + (Ax+B)\cos(x)e^x + A\sin(x)e^x + (Dx+D)\cos(x)e^x - (Dx+C)\sin(x)e^x + D\cos(x)e^x$

Substituting we get

$(Ax+B)\sin(x)e^x + (Ax+B)\cos(x)e^x + A\sin(x)e^x + (Dx+C)\cos(x)e^x - (Dx+C)\sin(x)e^x + D\cos(x)e^x - 2(Ax+B)\sin(x)e^x - 2(Dx+C)\cos(x)e^x = x\sin(x)e^x$

$\Rightarrow -(Ax+B)\sin(x)e^x + (Ax+B)\cos(x)e^x + A\sin(x)e^x - (Dx+C)\cos(x)e^x - (Dx+C)\sin(x)e^x + D\cos(x)e^x = x\sin x e^x$

$\Rightarrow -A-D = 1 \Rightarrow A = -1-D$

$-B+A-C = 0$

$A-D = 0 \Rightarrow A = D = -\frac{1}{2}$

$B-C+D = 0$

$-B-\frac{1}{2}-C = 0 \Rightarrow B = -\frac{1}{2}-C$

$B-C-\frac{1}{2} = 0$

sub. $-\frac{1}{2}-C-C-\frac{1}{2} = 0$

$\Rightarrow C = -\frac{1}{2} \Rightarrow B = 0$

$\therefore y_p = -\frac{1}{2}x\sin(x)e^x - \left(\frac{1}{2}x+\frac{1}{2}\right)\cos(x)e^x$

Example 6.

Find the general solution to

$$y'' + 9y = 2\cos 3x$$

Soln

The auxiliary equation is

$$m^2 + 9 = 0$$

$$\Rightarrow m = \pm 3i$$

$\therefore$ The complementary solution is

$$y_c = C_1 \sin 3x + C_2 \cos 3x$$

Let $y_p = Ax\cos 3x + Bx\sin 3x$

$$\Rightarrow y_p' = -3Ax\sin 3x + A\cos 3x + 3Bx\cos 3x + B\sin 3x$$

$$\Rightarrow y_p'' = -9Ax\cos 3x - 6A\sin 3x - 9Bx\sin 3x + 6B\cos 3x$$

Substituting into the equation we get

$$-9Ax\cos 3x - 6A\sin 3x - 9Bx\sin 3x + 6B\cos 3x + 9Ax\cos 3x + 9Bx\sin 3x = 2\cos 3x$$

$$\Rightarrow -6A = 0 \quad \text{and} \quad 6B = 2$$

$$\Rightarrow A = 0 \quad \text{and} \quad B = \frac{1}{3}$$

$\therefore$ The particular solution is

$$y_p = \frac{1}{3}x\sin 3x$$

The general solution is

$$y = y_c + y_p$$

$$= C_1 \sin 3x + C_2 \cos 3x + \frac{1}{3}x\sin 3x$$

Superposition Principle And Undetermined Coefficients

Let y_1 be a solution to

$$ay'' + by' + cy = f_1(x)$$

and y_2 be a solution to

$$ay'' + by' + cy = f_2(x)$$

Then $k_1y_1 + k_2y_2$ is a solution to

$$ay'' + by' + cy = k_1f_1(x) + k_2f_2(x)$$ where k_1 and k_2 are constants.

Example 1

Find the particular solution to

(a) $y'' + 2y' + 4y = x + \cos 2x$

(b) $y'' + 5y' + 6y = 6x^2 + 10x + 2 + 12e^x$

Soln

(a) Let $y_1 = Ax + B$ and $y_2 = C\cos 2x + B\sin 2x$

$\Rightarrow y_1' = A$ $\Rightarrow y_2' = -2C\sin 2x + 2B\cos 2x$

$\Rightarrow y_1'' = 0$ $\Rightarrow y_2'' = -4C\cos 2x - 4B\sin 2x$

For y_1,

$2A + 4Ax + 4B = x$

$\Rightarrow 4A = 1$ and $2A + 4B = 0$

$\Rightarrow A = \frac{1}{4}$ and $B = -\frac{1}{8}$

$\therefore y_1 = \frac{1}{4}x - \frac{1}{8}$

For y_2

$-4C\cos 2x - 4B\sin 2x - 4C\sin 2x + 4B\cos 2x + 4C\cos 2x + 4B\sin 2x = \cos 2x$

$\Rightarrow C = 0$ and $D = \frac{1}{4}$

$\therefore y_2 = \frac{1}{4}\sin 2x$

So $y_p = \frac{1}{4}x - \frac{1}{8} + \frac{1}{4}\sin 2x$

(b) Let $y_p = Ax^2 + Bx + C + De^x$

$\Rightarrow y_p' = 2Ax + B + De^x$

$\Rightarrow y_p'' = 2A + De^x$

Substituting

$$2A + 5e^x + 10Ax + 5B + 5De^x + 6Ax^2 + 6Bx + 6C + 6De^x = 6x^2 + 10x + 2 + 12e^x$$

So we have that

$A = 1$

$10A + 6B = 10$

$\Rightarrow B = 0$

$2A + 5B + 6C = 2$

$\Rightarrow C = 0$

and $D = 1$

$\therefore y_p = x^2 + e^x$